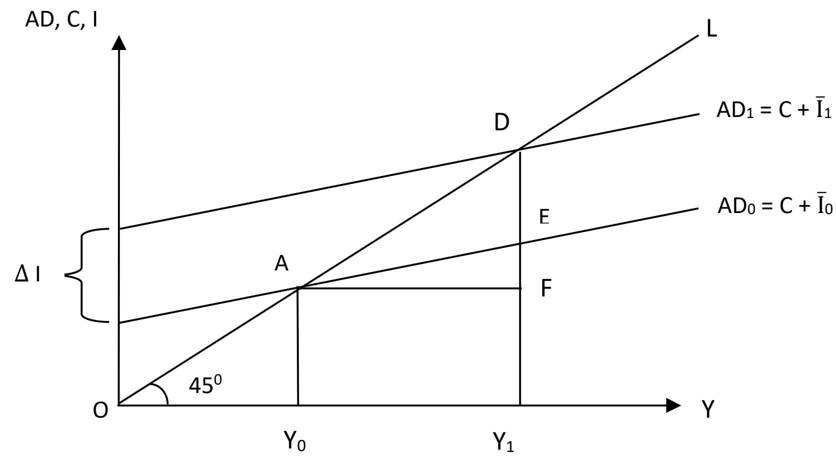

Fig. 2.5: Investment Multiplier

The initial equilibrium in Fig. 2.5 occurs at A where AD_0 (= C + $\bar{I}_0$) intersects OL. With an increase in autonomous investment to $\bar{I}_1$ (i.e., $\Delta I = \bar{I}_1 - I_0$), the AD curve shifts upward to AD_1 $(= C + \bar{I}_1)$ and the new equilibrium occurs at D where AD_1 intersects OL. Note that while investment increased by ΔI (= DE), the change in income ΔY (=DF) that it brings about is greater, i.e., $\Delta Y > \Delta I$ (where, $\Delta Y = Y_0Y_1$= AF = DF and DF > DE = ΔI), indicating the operation of a multiplier effect.

Induced Investment

Suppose the investment function is given by equation (2.2b) above, i.e., $I = c + dY$. In this case, investment demand has an induced component that is affected by changes in income. The equilibrium income would be derived as follows:

$Y = AD = a + bY + c + dY$, so that:

$$Y_e = (a + c) / [1 - (b+d)] \quad \ldots (2.7)$$

ceteris paribus, if there is a change in autonomous investment in this case (i.e., $\Delta I = \Delta c$; $\Delta a = 0$), the multiplier would be given by: $1 / [1 - (b+d)]$. This would follow since $\Delta Y = \{1 / [1 - (b+d)]\}\Delta c$. Note that in this case we assumed that $(b + d) < 1$ (i.e., the sum of MPC and MPI is less than one).

2.3.3 Government Expenditure and Tax Multipliers

We now consider the case of a three-sector economy. For simplicity we assume that investment demand is autonomous and the government incurs a constant amount of expenditure G which is autonomous. The government also collects taxes T. We will consider two cases: (i) the government collects a lump sum tax T, and (ii) the government sets a proportional tax rate t, so that its tax revenue $T = tY$. The government's budget deficit (B) is defined as $B = G - T - TR$, where TR refers to transfer payments made by the government. In what follows we will assume, for simplicity that $TR = 0$.

Lump Sum Taxes

When the government imposes a lump sum tax *T*, the equilibrium income would be given by (2.4) above: $Y_e = (a - bT + \bar{I} + G) / (1-b)$. In this case, the multiplier for any change in autonomous expenditure (either investment or government expenditure) would be given by: $1 / (1 - b)$. Thus, $\Delta Y/\Delta G = 1 / (1 - b)$ or $\Delta Y/\Delta I = 1 / (1 - b)$. If there is an increase in the amount of tax collected, then income would fall. In such a case the tax multiplier would be given by: $-b / (1 - b)$. Thus, $\Delta Y/\Delta T = - b / (1 - b)$. Recall that taxes are a leakage from the income stream, so increase in taxes leads to a decrease in disposable income. Consequently, there is a decrease in consumption demand, leading to a decline in output (as output is demand-determined, lower demand results in lowering of output). You should note that there is a two-way relationship: consumption is influenced by the level of income (evident from the consumption function) and income is influenced by consumption demand.

Balanced Budget Multiplier

Suppose the government increases its spending (i.e., $\Delta G > 0$) and cut taxes by exactly the same amount ($\Delta T < 0$). It implies that the government budget remains unchanged. If ΔB indicates budget balance, then $\Delta B = \Delta G - \Delta T = 0$. Alternatively we can write $\Delta G = \Delta T$. You may think that such action would have no impact on equilibrium income as the injection of additional demand (ΔG) is exactly offset by a leakage from the demand stream ($\Delta G = \Delta T$). However, we can show that the multiplier in this case is equal to one, i.e. income increases by the amount of increase in government expenditure. From equation (2.4) we know that $Y_e = (a - bT + \bar{I} + G) / (1-b)$, so that $\Delta Y = (-b\Delta T + \Delta G) / (1-b)$. Since $\Delta G = \Delta T$, this can be written as $\Delta Y = (1 - b)\Delta G / (1- b)$, or $\Delta Y / \Delta G = 1$.

Proportional Taxes

Now suppose the government levies a proportional tax on income at tax rate *t*, where $0 < t < 1$. Since $AD = a + b(Y - tY) + \bar{I} + G$, the equilibrium condition is given by

$$Y = a + b(1 - t)Y + + \bar{I} + G \qquad \dots (2.7)$$

By re-arranging terms, we obtain the equilibrium level of output as

$$Y_e = (a + \bar{I} + G) / \{1 - b(1 - t)\} \qquad \dots (2.8)$$

Equation (2.8) implies that the autonomous expenditure multiplier (for a change in investment or government expenditure) is now lower compared to the cases without the proportional tax. Without proportional taxes the multiplier is given by $1/(1 - b)$. Now, in the presence of proportional taxes, it is given by $\Delta Y / \Delta G = \Delta Y / \Delta I = 1 /\{1 - b(1 - t)\}$. The reduction in the value of the multiplier arises because, out of the increase in income in each round, a fraction *t* has to be paid in taxes.

You can see from equation (2.8) that any increase in the tax rate would lower equilibrium income, whereas any decrease in tax rate would have the opposite effect. Therefore, you should see the logic behind tax cuts offered by governments to revive economic activity. However, remember that our assumption is that there is excess production capacity, which is the main reason why the increase in consumption demand spurred by a cut in taxes would lead to an increase in output. Else, it will lead to price rise instead of increase in output.

2.3.4 Open Economy Multiplier

In an open economy there is an additional source of demand for domestic output, viz., exports (X). Exports are autonomous as they depend on foreign incomes (i.e., incomes of foreign residents who buy domestic products) rather than domestic income. Therefore, an increase in exports will have multiplier effect on income, via various rounds of induced increases in consumption. Firms and households purchase goods and services from foreign countries in an open economy. Such imports (M) are a leakage from the domestic expenditure and income streams. We assume that demand for imports depend on domestic income level. The import demand function is given by $M = h + mY$, where h is the autonomous component of import demand. Here, m is the *marginal propensity to import* (MPM) and $m > 0$. Note that both exports and imports are influenced by exchange rate changes also (we will take up such issues later in the course on International Economics). In the present Unit we consider any change in exports and imports due to exchange rate fluctuation is treated as an autonomous change.

In an open economy, exports are an injection while imports are a leakage from the stream of aggregate demand for domestic goods. Thus, aggregate demand is given by: $AD = C + \bar{I} + G + X - M = a + bY + \bar{I} + G + X - h - mY$. Therefore, the equilibrium condition is given by

$$Y = a + bY + \bar{I} + G + X - h - mY \qquad \ldots (2.9)$$

The equilibrium level of output or income is given by

$$Y_e = (a - h + \bar{I} + G + X) / (1 - b + m) \qquad \ldots (2.10)$$

In (2.10), the autonomous components are *I, G* and *X*. Any change in these autonomous components will have a multiplier effect. The autonomous expenditure multiplier for investment, government expenditure or exports is given by: $\Delta Y / \Delta G = \Delta Y / \Delta I = \Delta Y / \Delta X = 1 / (1 - b + m)$. Clearly, the open economy multiplier is smaller compared to the one for the closed economy (1/(1 – b). The reason is that a proportion m out of the increment in income is now spent on imports and domestic demand is lower to that extent as compared to a closed economy.

In general, from our discussion on multipliers you should note the following points: (i) the multiplier would be larger, the larger is the MPC (b) and MPI (d), the lower the tax rate (t) and the smaller the m; (ii) the multiplier is applicable both in case of a rise as well as a fall in autonomous incomes, meaning that income would fall by a multiple in case of a given cut in any autonomous component of demand; and (iii) the operation of the multiplier is driven by the assumption that the economy has ample excess capacity and unemployed resources, so that output can be demand-determined.

2.3.5 Limitations of the Keynesian Model

In the beginning of this section we mentioned that the Keynesian model is more relevant for an economy facing recessionary conditions, having excess production capacity in place. In economies having supply constraints (i.e., where there are supply bottlenecks), an increase in demand is likely to result in a rise in prices rather than in output. Second, Keynesian model would not apply to economies that are operating near full employment, where increase in output would require increase in factor and goods prices. Third, Keynesian model is generally used for macroeconomic analysis pertaining to the 'short run'. Normally, once wage contracts are written they last for some time; also, prices printed on sales catalogues are expected to be valid for some time. So the assumption of wage and price rigidity is quite reasonable for the short run.

Check Your Progress 2

1. State the assumptions on which the Keynesian model is based.

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2. You are given the following information for a closed economy: C = 10 + 0.75 Y and $\overline{I} = 20$.

   a) Find the aggregate demand function and calculate equilibrium income for this economy.

   b) Suppose autonomous investment increases to 25. How would equilibrium income change?

   c) What is the value of the multiplier?

   d) How would the value of the multiplier in (b) change if we introducethe government sector which imposes (i) only a lump sum tax, and (ii) only a proportional tax where $t = 0.1$?

   e) How would the value of the multiplier in (b) change if the economy is open and MPM = 0.3?

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3. By using appropriate diagram of the Keynesian model, explain how situations of disequilibrium (excess demand or excess supply) are self-correcting.

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2.4 AN ALTERNATIVE VIEW OF EQUILIBRIUM

The Keynesian model can alternatively be described through the saving-investment relationship. When saving is equal to planned investment, the economy is operating at the equilibrium level of output.

2.4.1 Saving Function

The saving function is a mirror image of the consumption function that gives the relationship between planned saving (S) and income (Y) in the economy. In the two-sector model, with only households and firms, aggregate income is either consumed or saved. Thus,

$$Y = C + S \qquad \dots (2.11)$$

The decision to consume more is essentially the same as the decision to save less.

The saving function may be derived as follows:

$$S = Y - (a + b\,Y) = -\,a + (1 - b)\,Y \qquad \dots (2.12).$$

Note that the intercept of the saving function is negative, indicating that 'dis-saving' (running down past saving) occur at relatively low levels of income. The slope of the saving function is $(1–b)$, which is known as the *marginal propensity to save* (MPS). Note that MPS = (1–MPC), so that higher the MPC, lower is the MPS. In economic analysis it is often assumed that richer households have higher MPS compared poor households.

In Fig. 2.6 we derive the saving function diagrammatically. The top panel represents the consumption function C, while the lower panel represents the saving function.

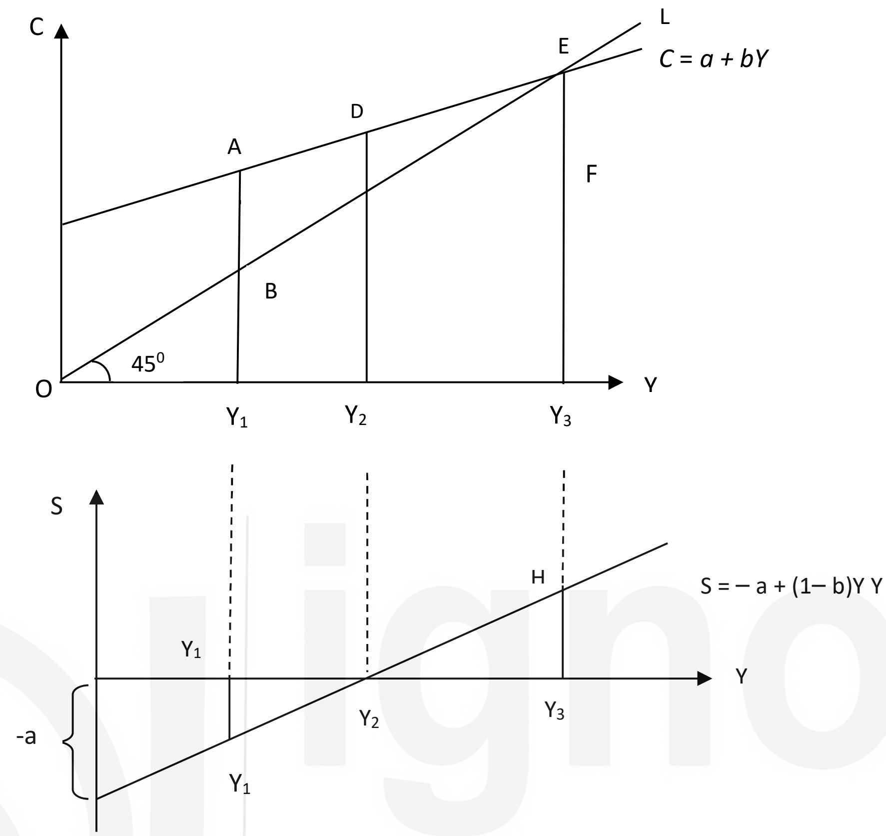

Fig. 2.6: Saving Function

At income level Y_2, where C intersects OL (the 45^0 line), planned consumption DY_2 is equal to income OY_2 (= DY_2) and planned saving is zero. At this level of income, the saving function intersects the x-axis. For all income levels less than OY_2 planned consumption exceeds income so that saving is negative. For example, in the top panel, at income level OY_1, planned consumption AY_1 is greater than income BY_1 (= OY_1) and the difference is reflected in negative saving Y_1G in the lower panel. For income levels such as OY_3, which is higher than OY_2, planned saving is positive (HY_3, in the lower panel).

2.4.2 Determination of Equilibrium Output

In a two-sector economy with only households and firms, equilibrium in the output market occurs at the level of income (or output) at which planned saving is equal to planned investment. This can be shown as follows: Suppose investment is autonomous and the investment function is given by $I = \overline{I}$ as in (2.2a). Recall that $Y = C + \overline{I}$ at equilibrium. We know that income is either consumed or saved, i.e., Y = C + S. Thus, we can put these together to write

$$C + S = Y = C + \overline{I}, \text{ or, } S = \overline{I} \qquad \dots (2.13)$$

In a closed economy without government, at equilibrium, planned saving must be equal to planned investment. Abstaining from current consumption (i.e., saving) involves producing fewer consumer goods and freeing resources for the production of investment goods. This can also be represented diagrammatically as in Fig. 2.7, where equilibrium income Y_e is attained at the intersection of the saving function and the investment function.

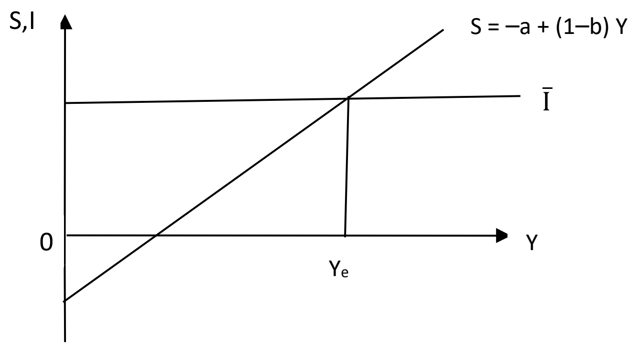

Fig. 2.7: Saving-Investment Equality

In a three-sector economy, the output market equilibrium condition is $Y = C + \bar{I} + G$. Income has three uses: consumption, saving and payment of taxes. Thus, $Y = C + S + T$. Putting these together, we can write the output market equilibrium condition as:

$$C + S + T = Y = C + \bar{I} + G, \text{ or, } S + T = \bar{I} + G \qquad \dots (2.14)$$

This can also be written as $S + (T - G) = \bar{I}$, which shows that at equilibrium in a closed economy, planned investment must be equal to total saving that include planned saving by households and public saving $(T - G)$. We can also re-write (2.14) as, $(S - \bar{I}) = (G - T)$. An implication of the above is: when planned investment is less than planned saving $(S - \bar{I} > 0)$, the government is running a budget deficit $(G - T > 0)$. The financing of the budget deficit is by borrowing from households.

2.4.3 Paradox of Thrift

An important result regarding the role of saving in the Keynesian model is referred to as the 'saving paradox' or the 'paradox of thrift'. You know that saving on the part of a household is considered as a virtue. Higher saving increases the assets and wealth of a household in the long run. For the economy as a whole, however, an increase in saving may not be good; it may be a vice. When households plan to save more, aggregate saving either remain the same or may even decrease. Let us explain it through a diagram (see Fig. 2.8).

In Fig. 2.8, equilibrium in the economy is at point A, where S = I. Output produced is Y_0. Suppose there is an increase in the autonomous component of households' planned saving, i.e., at each level of income planned saving is higher. This involves an upward shift of the saving function from S_0 to S_1. Note that S_1 is parallel to S_0 since there is no change in MPS. Now, the equilibrium output level increases to

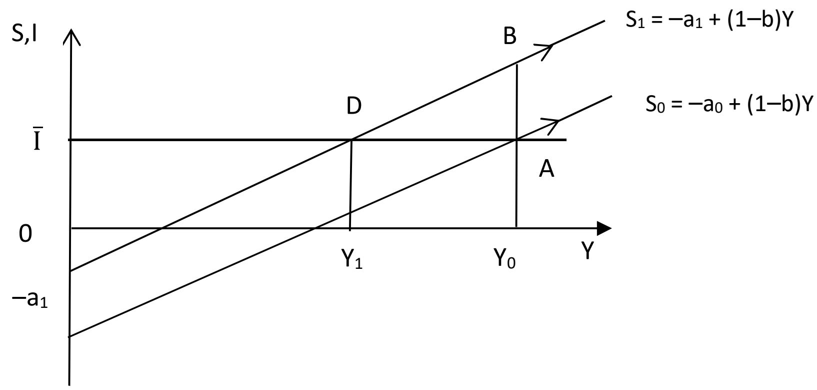

Fig. 2.8: Paradox of Thrift

Since there is no change in planned investment, there is disequilibrium in the economy now. Planned investment (DY_1) is the *same as before* ($DY_1 = AY_0$). Since people now wish to save more (saving function has shifted to S_1), income must be lower, so that saving is the same as before. The paradox is, despite an increase in thriftiness, there is no change in aggregate saving.

The desire to save more reduces planned consumption, thereby lowering aggregate demand and hence output (and income). At lower income levels, saving is correspondingly lower – instead of BY_0, aggregate saving fall to DY_1, since income has fallen to OY_1.

If the investment function has an induced component and is given by $I = c + d\,Y$, as in (2.2b), then an increase in planned saving can actually lead to lower level of saving at equilibrium, as shown in Fig. 2.9 below.

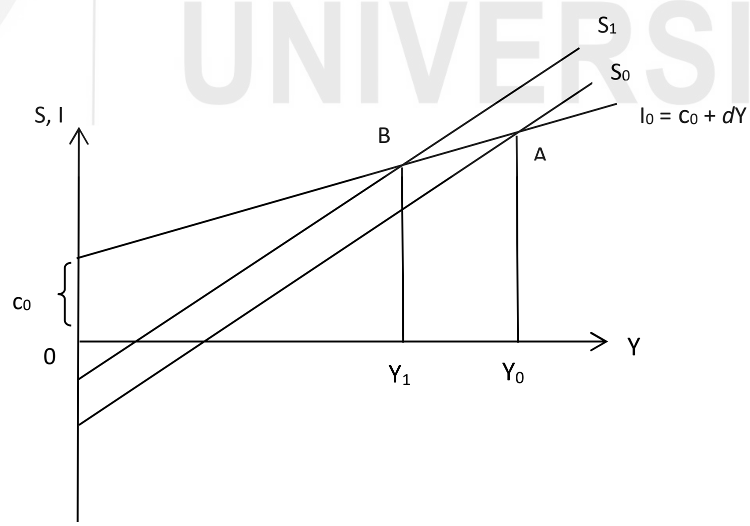

Fig. 2.9: Paradox of Thrift (Induced Investment)

In Fig. 2.9 the saving function shifts from S_0 to S_1. In this case, the reduction in consumption demand (due to rise in planned saving), leads to lower output which further leads to a decrease in induced *s*, thereby lowering income further. At the lower level of equilibrium income ($OY_1 < OY_0$), planned saving is correspondingly lower ($BY_1 < AY_0$), with the desire to save more leading to lower aggregate saving at equilibrium.

This result provides an important insight into the role of saving in the short run. In an economy that faces no supply constraints and has excess production capacity, an increase in the desire to save essentially lowers output by lowering aggregate demand. Therefore, at a lower equilibrium income level aggregate saving is the same or may even be lower than before. Note that such decline in output due to higher saving could be a short run phenomenon. In the long run, higher saving allows for higher investment and hence for creation of additional productive capacity. You will learn more about the role of saving in the long run while studying economic growth.

Check Your Progress 3

1. (a) In a two-sector economy (with only households and firms) derive the output market equilibrium condition in terms of planned saving and planned investment.

   (b) In a three-sector economy (with households, firms and government), what would be the relation between saving and investment at equilibrium if the government runs a budget surplus?

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2. What is meant by the 'paradox of thrift'? Use appropriate diagram to explain the cases of (i) investment demand is autonomous; and (ii) investment demand has an induced component.

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3. In a two-sector economy (with only households and firms), the consumption function is given by C = 10 + 0.75 Y and $\bar{I} = 20$.

(i) Derive the saving function.

(ii) Derive equilibrium output in terms of the equality of planned saving and investment.

(iii) Suppose there is an increase in the desire to save while planned investment remains unchanged. The new saving function is given by S = – 8 + 0.25 Y. Find out the new equilibrium income level.

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2.5 LIQUIDITY PREFERENCE

To complete our understanding of the Keynesian model we have to discuss the concept of 'liquidity preference' or the demand for money in this system. An important contribution of Keynes was to point out the interest sensitivity of the demand for money. Note that demand for money refers to the demand for 'real balances', i.e., nominal money demand deflated by the general price level. Since prices are constant in the Keynesian system, any change in money demand is essentially a change in the demand for real balances.

2.5.1 Components of Demand for Money

There are three main reasons people wish to hold money. The main components of the demand for money are (i) transactions demand, (ii) precautionary demand, and (iii) speculative demand. Let us discuss each of the above.

Transaction Demand: People hold cash since it is a medium of exchange and is useful for making transactions. This gives rise to transactions demand for money. You would have noticed that people with higher income tend to hold more money for transactions compared to the poor. So it is assumed that the amount of liquid cash people wish to hold for transactions is directly proportional to their incomes.

Precautionary Demand: People wish to hold money as it would be useful in contingencies such as medical emergency, accident, etc. It is reasonable to assume that precautionary demand for money is also an increasing function of income.

Speculative Demand: According to Keynes, people also hold money for speculation in the asset market, which gives rise to speculative demand for money. To understand why speculative demand for money is sensitive to the rate of interest, consider an economy where there are only two types of financial